

Lab Worksheet 2: Conditionals

CS 101 Algorithmic Problem Solving

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1. Where am I?

Chef is visiting Hunza and trying to find her way on the map. She is currently facing North. Each minute she turns exactly 90° in the clockwise direction.

There are only 4 directions: North, East, South, West (in clockwise order) and Chef wants to know the direction she will be facing after X minutes.

Given X , find the direction in which Chef is facing after exactly X minutes.

Constraints

- X
- $1 \leq X \leq 1000$

Interaction

The input contains a single line containing a single integer denoting the value of X .

The output must contain a single string—"North", "East", "South", or "West"—indicating the direction in which chef is facing after X minutes.

Sample

Input	Output
1	East
6	South

In the first case, $X = 1$. In 1 minute, Chef turns 90° clockwise once and ends up facing East.

In the second case, $X = 6$. In 6 minutes, Chef turns 90° clockwise 6 times, and ends up facing South.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
8	north
15	west
253	east

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
99	west
678	south
982	south

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input X; _____

Output; `d= X%4, if d==0 then print("north"), elif d==1 then print("east"), elif d==2 then print("south"), else print("west")`

2. Penalty Shots

In a football tournament, the final match has reached the penalty stage. Each team is given 5 shots at the goal and the team that scores a goal on the most number of shots wins the game. If both the teams' scores an equal number of goals, then the game is considered a draw and there will be 2 champions.

We are given ten integers $A, B, C, D, E, F, G, H, I, J$ where A, C, E, G, I values represent the goals of the first team and B, D, F, H, J values represent the goals of team 2. A value of 1 indicates that a goal was scored and a value of 0 indicates that a goal was not scored.

Determine the winner or find if the game ends in a draw.

Constraints

- $A, B, C, D, E, F, G, H, I, J \in \{0, 1\}$

Interaction

The input contains a single line containing 10 space-separated integers indicating the values of $A, B, C, D, E, F, G, H, I, J$.

The output must contain a single integer - 0 if the game ends in a draw, 1 if the first team wins, or 2 if the second team wins.

Sample

Input	Output
0 0 0 0 0 0 0 0 0 0	0
0 0 0 0 0 0 0 0 0 1	2

In the first case, $(A, B, C, D, E, F, G, H, I, J) = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$. No team scores any goal, so the game ends in a draw and 0 is the output.

In the second case, $(A, B, C, D, E, F, G, H, I, J) = (0, 0, 0, 0, 0, 0, 0, 0, 0, 1)$. The second team is able to score in their final shot, while the first team has scored no goals and hence the second team wins and 2 is the output.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1 0 1 0 0 0 0 0 0 0	1
1 1 1 1 1 1 1 1 1 0	1
1 1 0 1 0 1 0 1 1 0	2

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
1 0 1 0 1 1 1 0 0 1	1
0 1 0 1 0 0 0 0 1 1	2
1 1 0 0 1 1 0 0 1 1	0

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input; A, B, C, D, E, F, G, H, I, J let $q=A+C+E+G+I$ and $p=B+D+F+H+J$

Output; if $q > p$: print(1), elif $q < p$: print(2), else: print(0)

3. Single-Use Attack

Chef is playing a video game and is now fighting the final boss. The boss has H health points. Each attack of Chef reduces the health of the boss by X . Chef also has a special attack that can be used at most once, and decreases the health of the boss by Y . Chef wins when the health of the boss becomes zero or less.

What is the minimum number of attacks needed by Chef to win?

Constraints

- $a \in \mathbb{Z}$
- $Y \geq X$
- $0 \leq X, Y, H \leq 100$

Interaction

The input comprises a single line containing three space-separated integers denoting the values of H, X and Y .

The output must contain a single number denoting the minimum number of attacks needed by Chef to win.

Sample

Input	Output
100 25 40	4
46 1 2	45

In the first case, $(H, X, Y) = (100, 25, 40)$. One minimum sequence is to make 4 attacks of 25 each.

In the second case, $(H, X, Y) = (46, 1, 2)$. One minimum sequence is to make a special attack of 2 followed by 44 attacks of 1 each.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
100 29 45	3
78 15 78	1
0 30 45	0

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
45 8 12	6
99 6 41	11
100 3 6	33

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: H, X, Y

Output: if H=0: then A=0, elif H-Y<0 or H-X<0: THEN A=1,

else: A=ceil((H-Y)/X)+1 then print(A)

4. Figuring out the salary

After consultation with your financial advisor, you have devised the following payment scheme at your startup. An employee's *gross salary* is the sum of their basic salary, HRA, and DA. If the basic salary is less than \$1500, then HRA = 10% of the basic salary and DA = 90% of the basic salary. Otherwise, HRA = \$500 and DA = 98% of the basic salary.

Given S , the basic salary of an employee, as input, output their gross salary.

Constraints

- S
- $1 \leq S \leq 100000$

Interaction

The input comprises a single line containing the value of S .

The output must contain a single number upto 2 decimal places denoting the gross salary of the employee.

Sample

Input	Output
1203	2406.00
10042	20383.16

In the first case, $S = 1203$. The value of S is less than 1500 therefore, $HRA = 10\% = 120.3$, and $DA = 90\% = 1082.7$. Therefore the gross salary = $1203 + 120.3 + 1082.7 = 2406.00$

In the second case, $S = 10042$. The value of S is greater than 1500 therefore, $HRA = RS.500$, and $DA = 98\% = 9841.16$. Therefore the gross salary = $10042 + 500 + 9841.16 = 20383.16$

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1312	2624.00
2300	5054.00
1899	4260.02

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
1500	3470.00
1235	2470.00
7890	16122.20

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: S

Output: if $S < 1500$ then $A = S * 2$ print(A), else $A = S + (S * 0.98) + 500$

5. Game Winner

Nitin and Sobhagya were playing a game with coins. If Sobhagya has more coins then he is winning, otherwise Nitin is winning. Note that this means if both Nitin and Sobhagya have the same number of coins, then Nitin is winning.

Initially Nitin has A coins while Sobhagya has B coins. Then Ritik came and gave his C coins to the player who is not winning currently, after which Satyarth came and repeated the same process (gave his D coins to the player who is not winning currently).

Find the final winner of the game.

Constraints

- $1 \leq A, B, C, D \leq 1000$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of A, B, C and D respectively.

The output shall be S if Sobhagya is the final winner and N if Nitin is the final winner.

Sample

Input	Output
2 3 4 5	S
3 3 3 3	N

In the first case, Initially, Nitin has 2 coins and Sobhagya has 3 coins, so Sobhagya is winning. Then, Ritik gives his 4 coins to Nitin. Now Nitin has 6 coins and Sobhagya has 3 coins, so Nitin is winning. Then, Satyarth gives his 5 coins to Sobhagya. Finally Nitin has 6 coins and Sobhagya has 8 coins, so Sobhagya is the final winner.

In the second case, Initially, Nitin has 3 coins and Sobhagya has 3 coins, so Nitin is winning. Then, Ritik gives his 3 coins to Sobhagya. Now Nitin has 3 coins and Sobhagya has 6 coins, so Sobhagya is winning. Then, Satyarth gives his 3 coins to Nitin. Finally Nitin has 6 coins and Sobhagya has 6 coins, so Nitin is the final winner.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2 3 1 2	S
4 3 2 2	N
6 7 8 1	N

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
34 88 5 9	S
45 6 78 5	S
44 54 3 4	S

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: A, B, C, D

Output: if $A \geq B$: $B = (B + C)$ else: $A = (A + C)$
if $A \geq B$: $B = (B + D)$ else: $A = (A + D)$
if $B > A$: print("S")
else: print("N")